Jacob Sanders

Senior Forward Deployed Engineer

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SUMMARY

Applied AI engineer and two-time founder who has shipped production software end-to-end for 13+ years, from board bring-up to cloud-native SaaS. Works the way forward deployed engagements run: scope a vague problem directly with the people who have it, build and ship autonomously in weeks, then document it for whoever maintains it next. Ships production LLM applications on the Claude API, including multi-stage pipelines, structured extraction, semantic search, output guardrails, and a leak-free evaluation harness that scores live model calls against realized outcomes. Builds daily with Claude Code in a spec/plan/review loop and writes the repo context files that make coding agents execute reliably. Comfortable reading unfamiliar code and root-causing across the stack, with production experience in TypeScript, Python, C#, Go, and C/C++.

APPLIED AI ENGINEERING

- **Context engineering:** Builds daily with Claude Code in a structured spec/plan/review loop, maintaining repo context files, scoped task specs, and documented conventions so coding agents execute reliably across sessions
- **Evaluation harnesses:** Built a leak-free forward-tracking eval harness that logs every model call to an immutable ledger and scores live calls against realized outcomes (edge, hit-rate, Sharpe, t-stat), separating model quality from noise
- **Guardrails & generated-output safety:** Constrains model output with strict whitelists that prevent an LLM stage from overriding a deterministic ground-truth classification, and grounds a tutoring assistant so it cannot leak solutions; HMAC-verified webhooks, row-level access rules, and multi-tenant isolation on the data-handling side
- **LLM application delivery:** Ships production multi-stage LLM pipelines on the Claude API: structured JSON extraction, semantic search over embeddings, prompt caching, model tiering for cost (88% run-cost reduction), backoff retries, and Sentry observability
- **Customer-facing delivery:** Embeds with clients to scope from scratch and ship end-to-end in weeks, translating between non-technical stakeholders and the system being built

EXPERIENCE

Lien Into Revenue Services, LLC — Remote
Co-Founder / Lead Engineer — August 2025 – Present

- Co-founded and shipped a production SaaS platform end-to-end that manages the full personal-injury lien lifecycle for chiropractic practices, from patient intake through negotiation and settlement collection
- Scope features directly with the non-technical stakeholders who use the product, then own each one through implementation, release, and post-release support and triage
- Built the Next.js/React/TypeScript frontend on a serverless AWS backend (Lambda, AppSync GraphQL, DynamoDB, Cognito, S3) deployed as TypeScript infrastructure-as-code, with role-based multi-tenant access rules
- Ship under safe release practice: Jest and React Testing Library unit tests, Playwright end-to-end suites, code review, CI/CD, semantic versioning, and per-branch deploy environments so changes roll forward and back independently

ARK Electronics — Remote
Full-Stack Engineer (1099 Contract) — February 2025 – April 2025

- Ran the full engagement cycle as an embedded contractor: discovery and scoping from scratch with the client, then shipping a cloud-based manufacturing traceability platform end-to-end in weeks, covering product setup, serial number generation, barcode artifacts, and test-result tracking
- Built a React/TypeScript frontend and a serverless AWS backend exposing dual API surfaces, GraphQL for the web UI and REST for external clients, with API key management, usage plans, throttling, and event-driven status updates
- Translated shop-floor process from manufacturing and QA staff into working software, adjusting scope as requirements sharpened during the build

The Honeydew App, Inc. — Salt Lake City, UT
Founder — April 2023 – February 2025

- Founded an on-demand services marketplace; owned all system architecture and the majority of engineering while directing 4 University of Utah CS interns across frontend and backend workstreams
- Built a Flutter (MVVM) consumer application on a serverless TypeScript and Supabase/PostgreSQL backend, designing the relational schema, authentication, row-level access rules, and real-time job dispatch

Nerd United Lehi, UT (Hybrid)
Blockchain Engineer April 2022 – April 2023

- Built and deployed a production generative-AI pipeline in Python, orchestrating an image-generation API, IPFS pinning, and ERC-721 contract deployment as an automated multi-step workflow
- Developed applications in Go and Solidity on EVM-based protocols; deployed and operated 8 Proof-of-Authority test networks and 4 production networks

Skullcandy Park City, UT
Firmware Architect October 2018 – March 2022

- Led 3–5 engineers through the full product lifecycle from prototype to mass-production release, and served as primary liaison to Apple's MFi program office, translating technical constraints for executives, product managers, and non-technical partners
- Authored a C# desktop application for USB firmware updates and a Python CLI build toolchain with automated regression tests, replacing a manual process and compressing nightly validation cycles
- Established platform-wide design patterns and coding standards across a portfolio of products, judging when to build custom and when to extend an existing pattern for reuse
- Gated product features behind flags resolved from a hardware capability manifest, and routed firmware upgrades down separate tracks keyed on hardware ID and installed version to ship a non-backward-compatible hardware revision that broadcast the same device identifier

Code Corporation Draper, UT
Senior Firmware Engineer August 2013 – October 2018

- Led 3–4 engineers building advanced barcode reading systems from board bring-up through production release; drove Agile/Scrum adoption in the team
- Implemented a secure bootloader and BLE firmware for the Nordic nRF52832 SoC, contributed Linux kernel patches, and integrated a JavaScript engine into device firmware so customers could reconfigure behavior without a release

TECHNICAL SKILLS

AI / LLM	Claude API (Anthropic SDK), multi-stage LLM pipelines, context engineering (spec/plan/review, repo context files), evaluation harnesses & backtesting, structured outputs, semantic search (embeddings), guardrails & output whitelists, prompt caching, model tiering & cost optimization
Languages	TypeScript, Python, C#, Go, C, C++, Dart, Swift, Kotlin
Web / Mobile	Next.js, React, Flutter (MVVM), TypeScript, Vite, GraphQL, Figma
Cloud / Backend	AWS (Lambda, AppSync, API Gateway, DynamoDB, Cognito, S3, Amplify), Supabase/PostgreSQL, FastAPI, ASP.NET Core, REST & GraphQL API design, data modeling & schema design
Testing & Release	pytest, xUnit, Jest, React Testing Library, Playwright (E2E), LLM eval harnesses, CI/CD, feature flags, per-branch deploy environments, semantic versioning, code review, Agile/Scrum
Security & Data	HMAC webhook verification, Cognito auth, row-level access rules, multi-tenant isolation, API key management & throttling
Embedded / HW	BLE/Bluetooth, Apple MFi/iAP2, USB (Mass Storage, HID), SPI/I2C/UART, RTOS, bare-metal ARM

PROJECTS

Sentics: Multi-Stage LLM Signal Pipeline Python, Claude API, Next.js, Supabase/PostgreSQL
Serverless pipeline screening top crypto assets. A deterministic technical-analysis stage owns the directional call; a Claude stage writes rationale under a strict output whitelist that cannot override it; a cheaper stage distills news into structured sentiment. Model-tiered for cost (88% run-cost reduction), with a leak-free forward-tracking eval harness scoring live calls against realized returns.
github.com/jsanders5/sentics

PySando: Interactive Python Learning Platform FastAPI, React, Claude API, pytest
Runs learner code in a sandboxed subprocess and grades it against hidden pytest suites, paired with a Claude tutor grounded in the live lesson, exercise, and learner code, guardrailed to never reveal solutions. Async multi-turn state and provider abstraction.
github.com/jsanders5/PySando

EDUCATION

University of Utah Salt Lake City, UT
M.S., Electrical & Computer Engineering • B.S., Computer Engineering 2013